Smart Fridge MVP
Hardware Upload Requirements — Cooked Meals

For the hardware teammate • Device FRIDGE001 • Raw-data structure v1
Please provide the measurements and records below for cooked-meal trials. Cooked rice is the first example. This document covers your hardware and upload responsibilities only. You do not need to calculate food-spoilage decisions or percentages.

1. Required hardware data

- Upload both gas channels with the statistics already available: ppm estimate, sample count n, ADC rail hits, Rs/R0, and raw/resistance/voltage mean, min, max and SD.
- Identify the actual sensor model and intended gas for each channel. mq1/mq2 are channel names, not proof of sensor model or selective gas measurement.
- Provide firmware version, ADC maximum/reference, supply voltage, load resistances, R0 values, gas-curve coefficients and the actual sampling/record/upload periods in each run's metadata. State resistance units explicitly.
- Provide calibration state and profile/version, plus sensor quality and settling/saturation/noise flags. Use pending_validation or unknown when verification is incomplete.
- Record temperature/humidity only if measured. Do not invent or derive them from gas readings.

2. Required Firebase structure

Use these paths for new cooked-meal data. <...> means replace with an actual identifier; do not include angle brackets in the Firebase key.

```
experiments/<experimentId>/meta
experiments/<experimentId>/runs/<runId>
experiments/<experimentId>/observations/<observationId>
experiments/<experimentId>/events/<eventId>
telemetry/FRIDGE001/latest
telemetry/FRIDGE001/runs/<runId>/samples/<measurementId>
```

Write one experiment metadata record per physical meal sample. Write one run record per boot/continuous recording session. Store each measurement envelope under the raw history path and replace latest with the newest complete envelope. History may be uploaded in batches.
If you must keep the existing baseline/<runId>/samples/<epoch> path temporarily, preserve it and register samplesPath/metaPath in the experiment run record. Add experiment_id and

purpose=food_test to its run metadata. Tell us the exact paths so we can read them. Do not delete earlier data to adopt this structure.
Do not upload raw measurements to devices/FRIDGE001/latest or decisionHistory; those are reserved for assessments. Confirm producer write access before using new experiments/telemetry paths; the prepared read-rules patch does not grant those hardware writes. Provide the hardware Firebase user UID for producer authorization, not passwords or private service-account keys.

3. Experiment metadata — upload once per meal sample

Destination: experiments/COOKED-RICE-EXAMPLE-001/meta
All IDs, timestamps and location placeholders below are illustrative. Replace them with the actual cooked-meal trial. If a time or measurement is unknown, omit that optional value and note the uncertainty.

```
{
  "schemaVersion": 1,
  "experimentId": "COOKED-RICE-EXAMPLE-001",
  "deviceId": "FRIDGE001",
  "sampleId": "MEAL-SAMPLE-EXAMPLE-001",
  "food": {
    "name": "cooked rice",
    "category": "cooked_meal",
    "preparation": "cooked",
    "cookedAtMs": 1791954389000
  },
  "startedAtMs": 1791957989000,
  "storedAtMs": 1791957989000,
  "purpose": "food_test",
  "status": "running",
  "storage": {
    "location": "REPLACE_WITH_ACTUAL_TEST_LOCATION",
    "container": "REPLACE_WITH_ACTUAL_CONTAINER_DETAILS"
  }
}
```

Required: schemaVersion=1; experimentId; deviceId; sampleId; food.name; food.category=cooked_meal; food.preparation=cooked; purpose=food_test; actual startedAtMs; status; storage.location. Record cookedAtMs and storedAtMs whenever known. CookedAtMs is when cooking completed; storedAtMs is when this sample entered the reported storage location; startedAtMs is when this experiment began recording. These times are not interchangeable.
Record recipe/ingredients, portion amount with unit, container/open-or-covered condition and prior storage/reheating history when known. Update those details if conditions change. Status means running, interrupted, completed or abandoned; it is not a food-freshness label.

experimentId identifies the trial; sampleId identifies its physical cooked-food sample. Test one identified sample at a time. If multiple dishes share the sensing compartment, record all of them and do not attribute the gas signal to one item without evidence.

4. Raw measurement — upload repeatedly

Destination for current sample: telemetry/FRIDGE001/latest
Destination for the same envelope in history: telemetry/FRIDGE001/runs/RUN-EXAMPLE-001/samples/MEASUREMENT-EXAMPLE-001
The numeric gas values below are taken from the old clean-air export only to demonstrate the existing field format. They are NOT cooked-rice readings. Upload actual readings and actual times; do not reuse this example as experimental data.


```
{
  "schemaVersion": 1,
  "deviceId": "FRIDGE001",
  "experimentId": "COOKED-RICE-EXAMPLE-001",
  "sampleId": "MEAL-SAMPLE-EXAMPLE-001",
  "runId": "RUN-EXAMPLE-001",
  "measurementId": "MEASUREMENT-EXAMPLE-001",
  "measuredAtMs": 1791957989000,
  "uploadedAtMs": 1791958124000,
  "expectedUpdateSeconds": 60,
  "mode": "food_test",
  "calibration": {
    "state": "pending_validation",
    "version": "REPLACE_WITH_ACTUAL_PROFILE_VERSION"
  },
  "quality": { "state": "unknown" },
  "sample": {
    "epoch": 1791957989,
    "uptime_s": 2640,
    "mq1": {
      "n": 30, "ppm": 10.035, "rail": 0, "rs_r0": 2.55621,
      "raw": { "mean": 565.5, "min": 560, "max": 572, "sd": 3.14 },
      "rs": { "mean": 19944.2, "min": 19694.2, "max": 20159.1, "sd": 121.49 },
      "volt": { "mean": 0.4557, "min": 0.4513, "max": 0.461, "sd": 0.00253 }
    },
    "mq2": {
      "n": 30, "ppm": 30.007, "rail": 0, "rs_r0": 1.63274,
      "raw": { "mean": 1210.1, "min": 1196, "max": 1225, "sd": 6.15 },
      "rs": { "mean": 5654.4, "min": 5569, "max": 5737.2, "sd": 35.69 },
      "volt": { "mean": 0.9752, "min": 0.9638, "max": 0.9872, "sd": 0.00496 }
    }
  }
}
```

Required envelope fields are shown above. All outer timestamps are Unix milliseconds; the existing nested sample.epoch remains Unix seconds and must describe the same acquisition time. measurementId identifies one acquired reading; reuse it when retrying that reading. runId identifies its recording session. expectedUpdateSeconds states the actual latest-upload cadence.
Allowed mode: calibration or food_test. Calibration state: collecting_baseline, pending_validation or validated. Quality state: valid, settling, fault or unknown. Mark faulty/settling readings explicitly rather than removing them or treating their computed ppm as reliable. Add per-channel fault flags when available.
Keep historical sample fields such as uptime_s, free_heap and unsent if already collected. Do not remove the original raw/Rs/voltage statistics to keep only ppm. Register run metadata at experiments/<experimentId>/runs/<runId>, including device/experiment/sample IDs, firmware, boot/reset information, calibration parameters and history paths.

5. Upload and restart requirements

- Aim for one latest summary upload every 60 seconds if the hardware supports it. Report the actual cadence. Hourly history batches may continue, but they should not be the only current-data upload.
- Publish a complete latest envelope. Preserve original acquisition time on retries/backfills, and never overwrite a newer latest sample with an older buffered sample.
- After power loss with the SAME meal sample: retain experimentId, sampleId and original cooking/storage/start times; create a NEW runId when recording restarts.
- After replacing the food or starting a new trial: create a NEW experimentId and sampleId. Retain previous experiment/run records.
- Log known restarts, interruptions, storage-condition changes and sample replacement with observedAtMs, type, runId and notes. If an outage start/duration is unknown, state that instead of inventing it. Separate sensor-power interruption from refrigeration-power interruption when known.
- After reboot, use quality=settling or unknown until readings meet the calibration/stabilization procedure. Retain calibration profile/version and indicate whether it is still valid after the restart.

Event example — experiments/<experimentId>/events/<eventId>:

```
{
  "experimentId": "COOKED-RICE-EXAMPLE-001",
  "runId": "RUN-EXAMPLE-002",
  "type": "device_restart",
  "observedAtMs": 1791958124000,
  "notes": "REPLACE_WITH_KNOWN_INTERRUPTION_DETAILS"
}
```

6. Food observations and handoff checklist

Record independent observations of the cooked-meal sample, with timestamp, method/source and notes. Allowed labels: unknown, fresh, spoiled or inconclusive. Use unknown/inconclusive if the condition has not been independently established. Do not use the sensor's own prediction or elapsed hours as a reference food-condition label.
Observation example — experiments/<experimentId>/observations/<observationId>:

```
{
  "experimentId": "COOKED-RICE-EXAMPLE-001",
  "sampleId": "MEAL-SAMPLE-EXAMPLE-001",
  "observedAtMs": 1791958124000,
  "label": "unknown",
  "method": "operator_note",
  "notes": "REPLACE_WITH_ACTUAL_OBSERVATION"
}
```

Before handoff, please send:

- Current cooked-meal experimentId/sampleId, meal name, cooking/storage/start times when known and the actual storage conditions.
- The exact RTDB paths; one complete new measurement envelope; and its run/calibration metadata.
- Actual sensor models/channel interpretation, calibration procedure/profile and stabilization/quality criteria.
- Known power/restart notes and independent food-condition observations.
- Hardware Firebase user UID so new-path write permissions can be configured.

MVP scope is cooked meals. Future trials may include produce such as tomatoes using another food.category and a new experiment, but do not mix those trials or their calibration/reference data into the cooked-meal dataset.